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Numerical Study on the Vertical Water Exit of A Cylinder with Cavity

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Abstract

When a high-speed body with cavity passes through water-air free surface and exits water, its mechanical environment and dynamic characteristics change significantly due to the great difference in density and viscosity between water and air. With focusing on this problem, the Computational Fluid Dynamics (CFD) method is applied to perform numerical calculation on the process of this vapor-liquid-gas flow during the water exit of a high-speed cylinder, with the Volume of Fraction (VOF) multiphase flow interface-capturing techniques and the overset grid technology. After the verification and validation of the CFD model through mesh convergence study and a water-entry experiment, cavity evolution and flow characteristics including pressure and velocity distribution during the water exit are analyzed. The effects of different initial velocities on the pressure distribution and drag characteristics of the cylinder are investigated. Calculated results show that the cavity collapse during water exit causes strong pressure fluctuation on the cylinder; when the cylinder exits water enveloped in a supercavity, the pressure distribution on its wall surface and surrounding water region is relatively uniform, and the drag changes gently, and thus the cylinder has good motion stability.

Key words: water exit, cavity, CFD

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1 Introduction

Compared with surface and land-based weapons, underwater-launched weapons carried on underwater platforms are covert deterrent forces. However, different with the single-phase flow problem during moving, the passage of the underwater-launched body through the water surface involves water-air interface. Owing to the great difference in density and viscosity between water and air, the mechanical environment and dynamic characteristics will inevitably experience significant changes during the water-exit process. Therefore, the investigation of water exit is of great practical significance to the shape design and motion characteristics prediction of underwater-launched bodies.

As the water-exit process features are severe non-linearity, the solution of the coupling problem at the contact point between solid and liquid has always been a difficulty in the water-exit problem (Luo and Mao, 2004; Wang and Huang, 2018; Shen, 2013). At present, studies on water-exit hydrodynamics have been mainly conducted through simplified experiments and computational simulations. Shi et al. (2014) carried out the experiment of water exit with supercavity using a smart-designed experimental setup, and captured the supercavity collapse as the projectile left water at an initial velocity of 40–70 m/s with a high-speed camera. Chen (2015) successfully obtained the vehicle surface pressure and motion attitude parameters in the water-exit process through underwater experiment, and proposed a method to measure the surface load of vehicles. Lu et al. (2019) studied the water-exit process of two side-by-side axisymmetric bodies. The hydrodynamics and motion characteristics of the bodies were evaluated with the interaction of body-body and body-fluid taken into account. Zhao et al. (2016) studied the evolution of partial cavity during the oblique water-exit of slender bodies through experiment. Ren et al. (2021) employed new experimental devices to explore the effects of the Froude number and the cavitation number on the evolution and collapse of cavities. Liu et al. (2014) performed an experimental study on the vertical water-exit under ventilated cavitation of axisymmetric bodies with conical head shape, revealing the phenomenon of ventilated cavitating flow during water exit. Truscott et al. (2016) captured the process of water exit and free-fall water entry of spheres with a high-speed camera and analyzed their motion characteristics. Piro and Maki (2012) conducted experiments and simulations to predict the vertical force generated by the water exit of a wedge-shaped body at a constant speed.

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Takamure and Uchiyama (2020) used a smart underwater launcher to study the change of the water-exit process with different sphere densities. It was found that the characteristics of the water layer of the free surface surrounding the sphere is the main factor affecting the flow energy during the water-exit process.

As a powerful supplement to experimental methods, computational fluid dynamics (CFD) can provide more detailed data for the analysis of flow field, forces and motion of a body during water exit. Li et al. (2007) numerically simulated the complex unsteady flow during the water exit of a slender axisymmetric body by the finite volume method, captured the water-air interface by a VOF method, and analyzed the change of forces acting on the body during water-exit. Tan et al. (2019) numerically simulated the water-exit process of a morphing cross-media aerial underwater vehicle under typical working conditions, and analyzed the transient properties of the flow field and load distribution of the vehicle at different stages of the water-exit process. Min et al. (2010) studied the cavitating multiphase flow and hydrodynamic evolution of an underwater-launched missile in vertical launch process by numerical simulations. The motion of the missile was investigated with the influence of gravity and the effects of the missile head shape being discussed. Liu et al. (2005) successfully simulated the generation, expansion, contraction and separation of the trailing cavity for an underwater-launched missile during free flight, and obtained the change laws of surrounding velocity field and pressure field. Pei et al. (2017) established a quasi three-dimension hydrodynamic model of cavity collapse for the water layer with certain thicknesses, and analyzed cavity evolutions of an underwater vehicle during water exit. Chen (2012) modified the turbulence model to accurately describe the viscous flow separation in the water-exit process, and numerically investigated the high-speed water-exit problems for axisymmetric bodies with flat head and hemispheroid head respectively. Wang et al. (2018) simulated the trajectory of an unmanned underwater vehicle launching from submarine and analyzed the pressure of flow field during the whole launching process. Hao et al. (2019) numerically simulated the vertical water-exit of spheres at different velocities using the Lattice Boltzmann Method (LBM) method, and studied the peak value, free surface breaking, wake field and hydrodynamic force acting on the sphere. Ni et al. (2015) performed calculation for the water-exit process of a fully submerged body using finite element software, and proposed a numerical model for a fully submerged object penetrating the free surface. Tassin et al. (2013) proposed an analytical model applicable to describing the water entry and exit of a body whose shape varies with time. Korobkin (2013) put forward a model of hydrodynamic loads acting on the floating rigid body during the lifting of the body from the liquid surface. Xu et al. (2019) used Star-CCM+ to study the characteristics of flow around projectiles successively being launched underwater and exiting from the water, and found that there are two typical wake vortex structures as the angle of attack of water exit changes.

Although many researches have been done in the field of water exit, more in-depth researches are needed to study the complex multiphase flow characteristics of the water exit process of high-speed bodies with cavities and the characteristics of forces acting on the body. When the high-speed body with cavities passes through the free surface squeezed and disturbed by the free surface, the closed cavity attached to the body may collapse or open and connect with the atmosphere, thus changing the flow environment around the body from water and vapor to a more complex combination of water, vapor and air. In this process, there is a strong interaction between the vapor-liquid-gas multiphase flow field and the body. Figuring out this interaction mechanism can provide valuable guidance for solving the mechanical problems in water exit, which is of great interest in the engineering field. In view of this, a CFD method is applied in this paper to study the process of a cylinder with cavity crossing the free surface under different cavitation conditions, and the cavity evolution, the force on the cylinder and the flow field variation during the water exit is detailed and discussed. The study tries to provide a valuable reference for the structural design and motion stability analysis of water-exit cylinders.

2 CFD modeling

2.1 Numerical model and boundary conditions

In the water-exit process of a high-speed body, cavitation may occur with cavities forming partially or totally enveloping the whole body, and this is a vapor-liquid-gas multiphase flow process. In the present study, the VOF and Zwart−Gerber−Belamri methods are used together to construct the numerical configuration for the high-speed water-exit process.

The basic equation of the numerical model can be expressed as follows.

(1) Continuity equation:

$$\frac{\partial \rho}{\partial t}+\nabla(\rho \boldsymbol{u})=0. \tag{1}$$

(2) Momentum equation:

$$\frac{\partial}{\partial t}(\rho \boldsymbol{u})+\nabla\cdot(\rho \boldsymbol{u}\boldsymbol{u})=-\nabla p+\nabla\cdot\left\{\mu\left[\left(\nabla \boldsymbol{u}+\nabla \boldsymbol{u}^{\mathrm{T}}\right)-\frac{2}{3}\nabla\cdot \boldsymbol{u}\boldsymbol{I}\right]\right\}. \tag{2}$$

(3) Air-phase volume fraction transport equation:

$$\frac{\partial}{\partial t}(\alpha_{\mathrm{a}}\rho_{\mathrm{a}})+\nabla(\alpha_{\mathrm{a}}\rho_{\mathrm{a}}\boldsymbol{u})=0. \tag{3}$$

(4) Vapor-phase mass transfer equation:

$$\frac{\partial}{\partial t}(\alpha_{\mathrm{v}}\rho_{\mathrm{v}})+\nabla(\alpha_{\mathrm{v}}\rho_{\mathrm{v}}\boldsymbol{u})=R_{\mathrm{e}}-R_{\mathrm{c}}, \tag{4}$$

where ρ and μ represent the density and viscosity coefficients

of the fluid respectively; u is the flow velocity; t represents time; the subscripts a, v and w indicate the air phase, vapor phase and liquid phase, respectively; R_e and R_c are the source terms of gasification and condensation, respectively. Their relationship is as follows:

$$\rho = \alpha_a\rho_a + \alpha_v\rho_v + (1-\alpha_a-\alpha_v)\rho_w; \tag{5}$$

$$\mu = \alpha_a\mu_a + \alpha_v\mu_v + (1-\alpha_a-\alpha_v)\mu_w; \tag{6}$$

$$R_e = F_{vap}\frac{3\alpha_{nuc}(1-\alpha_v)\rho_v}{R_B}\sqrt{\frac{2}{3}\frac{p_v-p}{\rho_w}}; \tag{7}$$

$$R_c = F_{cond}\frac{3\alpha_v\rho_v}{R_B}\sqrt{\frac{2}{3}\frac{p_v-p}{\rho_w}}, \tag{8}$$

where the coefficients $F_{vap}=50$, $F_{cond}=0.01$, $R_B=10^{-6}$m, $\alpha_{nuc}=5\times10^{-4}$ and p_v is the saturation vapor pressure.

The SST k–ω model is used to establish the transport equations of turbulence kinetic energy and dissipation rate, and the standard wall functions are adopted in the paper. The pressure-based solver is used for numerical calculation, and the Coupled method for the solution of pressure-velocity coupling. In the discretization scheme, the PRESTO! scheme is used for modeling of the pressure term, and the first-order upwind scheme for the density term, momentum term and energy term.

A model of an uniform steel cylinder is employed in the present study, with the cylinder length L=50.75 mm, diameter D=14.5 mm and weight m=65 g. The calculation domain of water exit is established as shown in Fig. 1. The cylinder moves vertically upwards at a certain speed (gravitational acceleration g=9.81 m/s^2, vertically downward) with the underwater position where the distance between the head and the free surface is the length of the cylinder. The surface of the cylinder, the bottom and four sides of the calculation domain are set as walls, and the upper part of the calculation domain is set as pressure-inlet. Unstructured grids are used to discretize the water region around the cylinder, and 6-DOF (degree of freedom) overset grid technology is used to assemble the subdomains and background domain. To better calculate the wave motion process of the cylinder passing through the free surface, the meshes around the still water surface are refined, and the time step is set as 10^{-5} s to ensure the calculation accuracy.

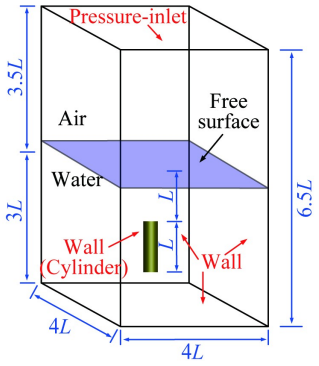


Fig. 1. Calculation domain and boundary conditions.

2.2 Mesh verification and model validation

Generally speaking, as the mesh density increases, the calculation accuracy improves, but meanwhile the calculation consumption also goes up because of the need for more computer resource and CPU time. In order to identify the effect of mesh number on the calculation of the water-exit process, and to find out a compromise grid scheme with high calculation accuracy and low calculation consumption, three sets of meshes, namely, coarse (M1, mesh number = 4.89 million), medium (M2, mesh number = 5.16 million) and fine (M3, mesh number = 5.54 million) meshes are proposed for the calculation model. The drag of the cylinder and the evolution of surrounding cavities during water exit are calculated when the initial velocity of cylinder is 30 m/s, as shown in Fig. 2. Water-exit drag coefficient of the cylinder is defined as $C_d=F/(0.5\rho V^2S)$, where F is the drag force during water exit, ρ=998.2 kg/m^3 is the water density, V is the initial exit velocity, and S=0.000165 m^2 is the cross-section area of the cylinder. As can be seen, when the number of meshes increases from M1 to M2, the cavity shape at the time corresponding to the drag force peak changes obviously, indicating that the M1 grid scheme cannot well capture the details of cavity evolution and drag force changes during water exit. And when the mesh number further increases to M3, the cavity shape at the time corresponding to the drag peak does not change significantly, indicating that the M2 grid scheme has been in a mesh-convergence state and further increase of the mesh number does not improve the results significantly. In addition, a comparison of calculation results is made under the conditions of 15 m/s and 45 m/s initial water-exit velocities, showing that the M2 grid scheme is quite acceptable as well. Therefore, the M2 grid scheme is employed for the calculation and analysis in subsequent investigation in this paper, and the corresponding mesh details are shown in Fig. 3.

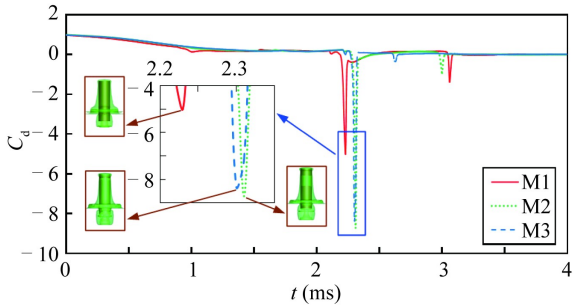


Fig. 2. Mesh verification results.

In the existing experimental conditions for the water exit of high-speed cylinder, high-pressure gas is usually used to push the cylinder out of the launch tube. The gas source for

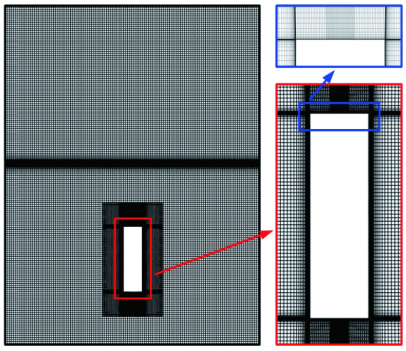


Fig. 3. Computational domain and mesh details around the cylinder.

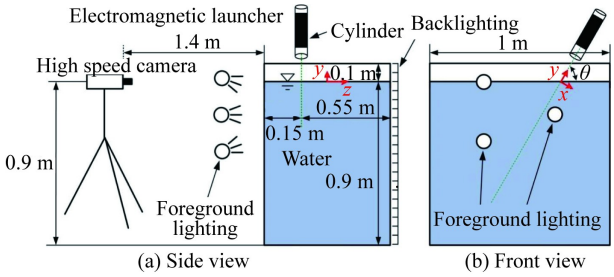


Fig. 4. Sketch of the experimental setup for water entry.

launch also affects the formation of cavities. This is quite different from the natural cavity calculated in this paper. At present, it is difficult to obtain experimental data of natural cavity in water exit. The water entry process of a cylinder is also a cross-media multiphase flow problem, accompanied by the cavitation process from water to vapor. Therefore, a water-entry experiment of a similar cylinder is used to verify the reliability of the CFD model in this study. As shown in Fig. 4, the water-entry experiment is carried out in a transparent water tank located in a closed laboratory, with a smart designed electromagnetic launcher being applied to release a cylinder (L/D=3.5) into water, similar to that detailed by Lyu et al. (2021, 2022). A high-speed camera is used to capture the water entry process and the velocity is calculated by the deviation of poisons of the cylinder between frames. Fig. 5 shows the comparison between the results from experiment and numerical calculation on oblique water entry of the cylinder with water-entry angel of 60° and velocity of 40.3 m/s. The evolution process of the cavity demonstrates that the numerical results are in a good agreement with those from experiments. What is more, it is found that the numerical results with cavitation model agree better with experimental results than those without cavitation model. Therefore, it is believed that the cross-media water-exit model employed in this paper is accurate and reliable.

3 Results and discussion

To analyze the effects of different launch velocities on the cylinder passing through free surface, the vertical water-exit process of the cylinder at three different initial launch velocities of 15, 30 and 45 m/s is studied, with the same launch depth, as shown in Fig. 1. The state when the underwater cylinder moves near the quiescent water surface but without obvious disturbance to the surface is defined as the initial state of the cylinder's water exit process, corresponding to t=0. The cavity evolution, surface pressure characteristics and drag characteristics of the cylinder during the whole water-exit process from the state of initial time to that of completely away from water are analyzed in the following parts.

3.1 A typical case for water exit (V=30 m/s)

Take the underwater initial velocity of the cylinder V=30 m/s as a typical example, the cavity evolution, pressure field and velocity field during the cylinder's water exit are analyzed. The evolution process of the cavity is shown in Fig. 6. At t=0 ms, the cavity has developed to a certain extent. At this time, the shoulder cavity covers the front half of the cylinder, and the tail cavity has a tendency to shrink. When the cylinder moves upwards, the squeeze between the cylinder and the free surface continuously lifts up the free surface, resulting in a bulge of the free surface as shown in Fig. 6b. In this process, the shoulder cavity continues to develop and gradually stabilizes, which is not greatly affected by the free surface, and the tail cavity has obvious shrinkage and re-entrant jet. As the cylinder starts to cross free surface, the free surface squeezes the cavity downwards. The local enlarged details for Figs. 6b–6c indicate that the front end of the cavity has begun to shrink. As the cylinder continues moving upwards, the shoulder cavity slides downwards under the continuous action by free surface. In this process, the front end of the shoulder cavity begins to collapse

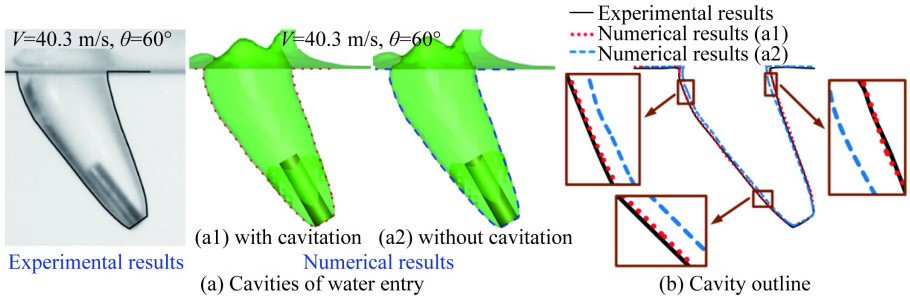


Fig. 5. Comparison of results between simulation and experiments.

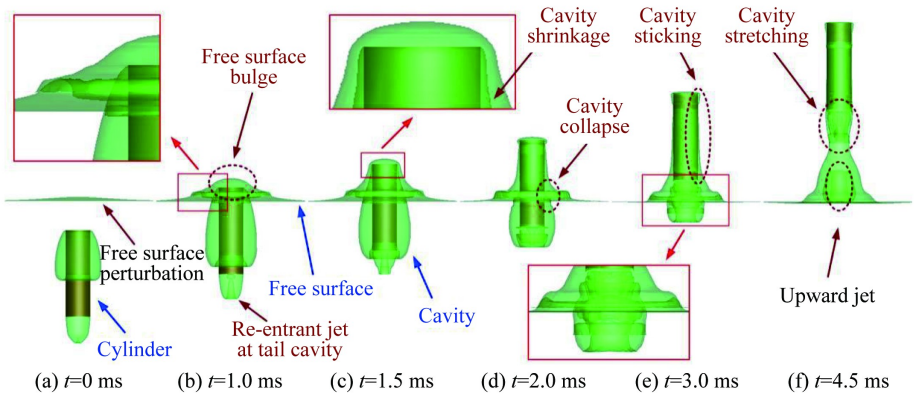


Fig. 6. Cavity evolution during cylinder's water exit (V=30 m/s).

and its rear end contacts and merges with the tail cavity. When the cylinder completely passes through free surface, the shoulder cavity moves downwards along the axis, the front end of the cavity sticks to the surface of the cylinder as shown in Fig. 6e, and the tail cavity collapses. Finally, cavity stretching occurs as shown in Fig. 6f, and the cavity near the free surface still accompanies the upward jet that remains developing.

During the movement of the cylinder in water-air flow, flow fields around the head and the tail cavity experience complex pressure changes with the pressure details around the cylinder as shown in Fig. 7. A high-pressure region is formed above the head of the cylinder due to the disturbance of the flow field by the dynamic pressure of the moving cylinder, as shown in Fig. 7a. As the cylinder continues moving upwards near the free surface, the water layer in the region above the head becomes thinner, and both the pressure at the head and the high-pressure range decrease significantly. The movement of cylinder and collapse of tail cavity result in concentration of flow to the tail end of cavity, as detailed for Fig. 7a. The existence of cavity boundary blocks the flow, leading to a high stagnation pressure region according to Bernoulli principle, as shown in Figs. 7a and 7b. The significant pressure difference inside and outside the cavity further causes the tail cavity re-entrant jet as shown in Fig. 6. Obviously, the cavity regions at the shoulder and the tail of the cylinder are both low-pressure regions. From t=0 ms to t=1.0 ms, both the low-pressure region and the cavity contour expand significantly; whereas, both the low pressure region and the cavity contour begin to shrink at t=1.5 ms (Fig. 7c). Two interfaces play an important role during the cylinder passing through the free surface (Wang et al., 2012). Interface 1 is the interface between water and external air (free surface), and Interface 2 is that between water and vapor in the cavity. Cavitation in the water-exit process is continuously affected by the two interfaces, which increases the surrounding pressure field and leads to cavity condensation and collapse. Collapse first occurs at the front end of the cavity (Fig. 7d), and advances downwards along the axis of the cylinder. There is also a high-pressure region at the tail, and the lower end of the cavity collapses upwards. When the cylinder moves across the air-water interface with the tail leaving water, the collapse of cavity is enhanced by the air-water interface breaking up, and leads to an enhancement to the high pressure region behind the tail of the cylinder (Fig. 7e). As the cylinder continues moving upwards to the height of 1 time its length, there is only a weak pressure gradient at the tail where cavity stretching occurs (Fig. 7f), the influence of the two interfaces on the motion of the cylinder gradually disappears, and only the aerodynamic influence needs to be considered in the hydrodynamic force of the cylinder.

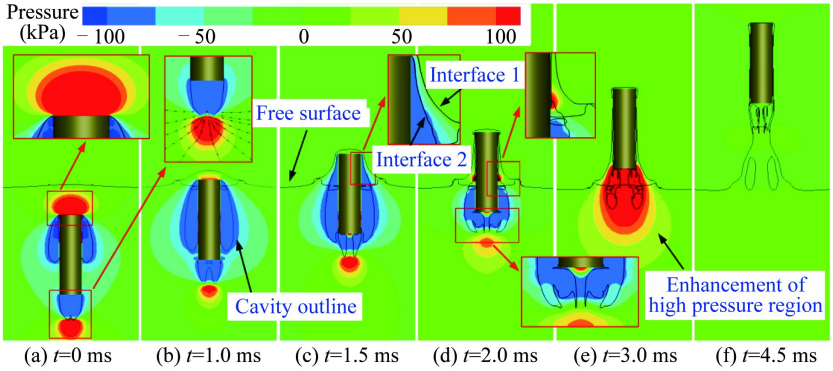


Fig. 7. Pressure variation of the flow surrounding the cylinder during water exit (V=30 m/s).

During the water exit of the cylinder, the velocity change of water region surrounding the cylinder is shown in Fig. 8. Since the squeezing by motion and the viscosity of the fluid, the cylinder will disturb the surrounding flow field, and a flow layer exists on the surface of its head and shoulder, in which the speed is basically equal to the speed of the cylinder (Fig. 8a). In addition, when the cylinder moves upwards, the incoming flow is blocked by the cavity. As a result, the water region along the cylinder and behind the shoulder cavity is not affected obviously. The flow region is only influenced by the shear stress of the cylinder wall with a thin boundary layer near the wall, and the flow region outside presents an undisturbed low velocity region, as detailed in Fig. 8a. As shown in Figs. 8b−8f, the upward motion of the cylinder produces an obvious semicircular hump near the air-water surface and the kinetic energy of the cylinder is transferred to the hump and then to air flow above, leading to a large block of low velocity field at the top of the free liquid surface. As a complex velocity gradient exists at the tail due to the tail re-entrant jet, there is a re-entrant flow zone of high-speed. At t=0 ms, the maximum velocity region is at the end of the cavity (Fig. 8a), and then develops to the cylinder tail and moves upwards until the cylinder completely exits the water (Fig. 8f). The wake region is the main dissipation region for the cylinder to transfer energy to the fluid. It can be found that with the upward movement of the cylinder, the wake region continues to expand, but the velocity distribution gradient tends to gradually flatten or even disappear as the cylinder moves away, and its main influenced region is the cavity shedding region directly behind the cylinder.

3.2 Effect of initial velocity on water exit

Based on the investigation of the water exit process at the same velocity, the impact of the initial underwater launch velocity on this process is further studied. With three initial velocities, i.e. 15 m/s, 30 m/s and 45 m/s being taken into account, the cavity shape and pressure distribution at the moments when the cylinder head crosses the free surface and when the cylinder completely exits the free surface are shown in Fig. 9, respectively. At the moment when the cylinder head crosses the free surface at different initial velocities, the pressure above the head changes obviously. The larger the initial velocity, the greater the dynamic pressure. This results in a more obvious high-pressure region in the water above the head, as shown in Fig. 9a. As demonstrated in the cavity shape diagram, when the initial velocity is V=15 m/s, the shoulder's and tail's cavities of the cylinder are small and unstable, and are prone to shed or collapse (Fig. 9a1). At high velocities, the cavities generated are rela-

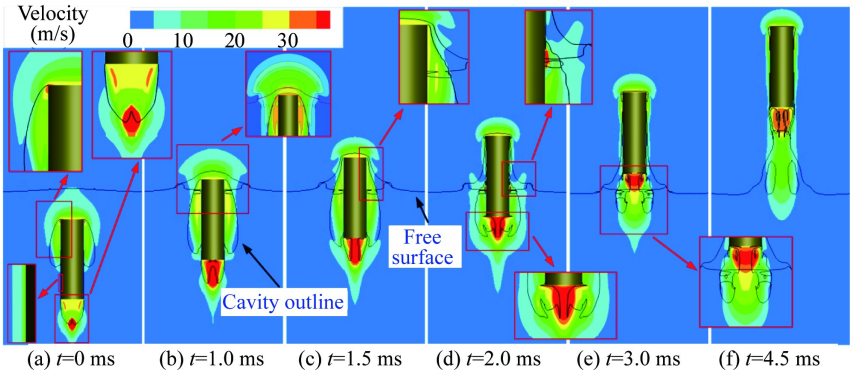


Fig. 8. Velocity variation of the flow surrounding the cylinder during water exit (V=30 m/s).

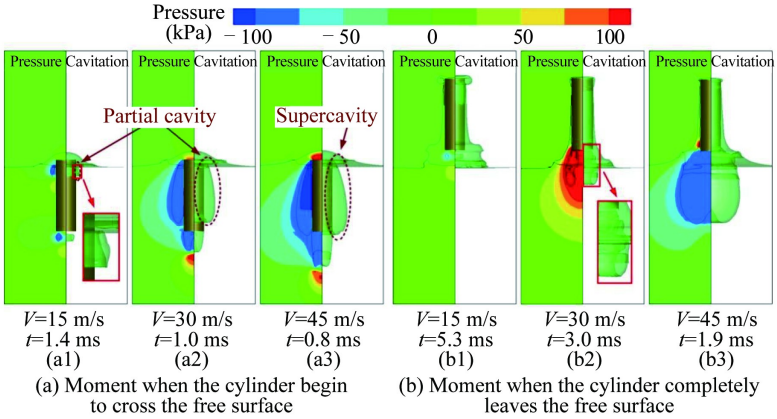


Fig. 9. Flow pressure and cavity evolution when the cylinder contacts the free surface and passes through the free surface.

tively stable, and the cavity covers most of the cylinder surface at V=30 m/s (Fig. 9a2); a supercavity is formed and completely envelops the cylinder surface at V=45 m/s (Fig. 9a3). Obviously, the cavitation becomes more distinct as the initial velocity of the cylinder increases, and the diameter and length of the cavitation generated during cylinder motion become much larger. Fig. 9b shows the comparisons of pressure distribution and cavity shape at the moment when the cylinder completely exits free surface at different initial velocities. It is found that the cavity of the cylinder exiting the water has basically collapsed and only a small part remains at the tail at V=15 m/s (Fig. 9b1). At V=30 m/s, the cavity of the cylinder exiting the water falls off from the tail and collapses, leading to an enhancement to the high pressure region behind the tail of the cylinder (Fig. 9b2)), as discussed above. The cavity of the cylinder exiting the water has certain stability at V=45 m/s. Only the front end of the cavity at the local high-pressure region collapses, and the lower end of the cavity is depressed toward the axis (Fig. 9b3)).

For the moment when the cylinder head crosses the free surface, the cylinder surface pressure distribution along the axial direction is shown in Fig. 10, in which x=0 corresponds to the cylinder head and x=1 to the tail. At V=15 m/s, the pressure distribution on the surface of the cylinder is relatively uniform. It is found that there is a relatively high-pressure region at 0.23L and the peak pressure reaches 150 kPa. At V= 30 m/s and 45 m/s, in the region where the cylinder is covered by the cavity, the surface pressure distribution is relatively uniform and the pressure value is small with the pressure increasing gradually from head to tail; in the region where the cavity covers the edge, due to the instability at the lower end of the cavity edge, the load acting on the cylinder increases and high pressure is formed, and the maximum pressure of the cylinder at high velocities is also very high. The peak pressure at 0.87L reaches 300 kPa for V=30 m/s and reaches 450 kPa at 0.99L for V=45 m/s. It should be noted that the formation of the high-pressure region and transient peak value mainly comes from the instantaneous impact caused by cavity collapse. The shoulder cavity is unstable and usually collapses at the end, leading to obvious flow impact on the side of cylinder. As shown in Fig. 11,

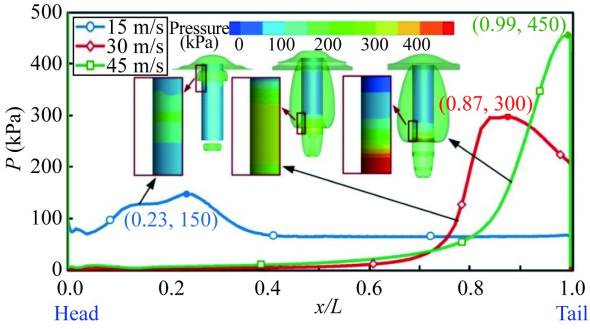


Fig. 10. Surface pressure along the cylinder with its head crossing the free surface.

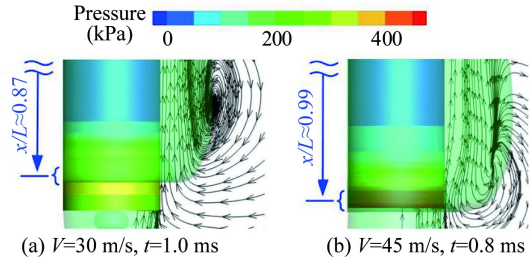


Fig. 11. Flow details at the end of shoulder cavity around the cylinder.

the flow jets onto the side wall of the cylinder at the end of the shoulder cavity at x/L≈0.87 of the cylinder for V=30 m/s and x/L≈0.99 for V=45 m/s, according with the pressure peak depicted in Fig. 10.

The distribution of surface pressure along the axis at the moment when the cylinder completely crosses the free surface is shown in Fig. 12. It is found that the cavity has fallen off completely and the cylinder is basically wetted at V=15 m/s. What is more, the surface pressure distribution of the cylinder is relatively uniform with a value about 110 kPa, while a partial cavity attaching on the head makes the surface pressure slightly lower. At V=30 m/s, the cavity is shedding off as the part of it remains attached to the cylinder surface. The pressure distribution fluctuates along the cylinder surface with the peak value reaching 240 kPa at 0.04L. As the initial velocity increases to 45 m/s, the cavity drops along the axis and merges with the tail cavity, making a low pressure region at the lower half of the cylinder. At this moment, the free surface slides downwards and squeezes the sticking part. The cavity at the cylinder head and shoulder interflows with the air and gets collapsed. As a result, the pressure fluctuates significantly with a peak value of 450 kPa at 0.11L.

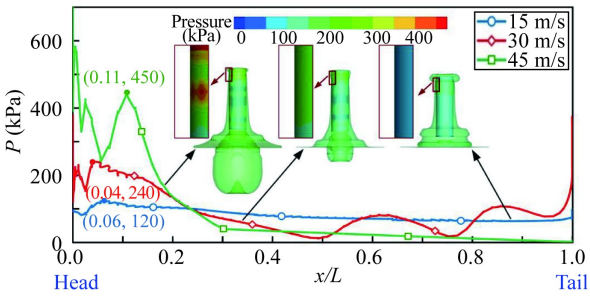


Fig. 12. Surface pressure along the cylinder when it completely crosses the free surface.

The drag coefficient variations of the cylinder during water exit at different velocities are shown in Fig. 13. Under the present water exit conditions, the drag coefficient C_d of the cylinder at different velocities decreases with time increase generally. The lower the initial velocity, the greater the drag coefficient and the slower the decrease. As marked in the figure, significant negative peaks of the drag coefficient (A1, A2, B1, B2) appear under partial cavitating conditions,

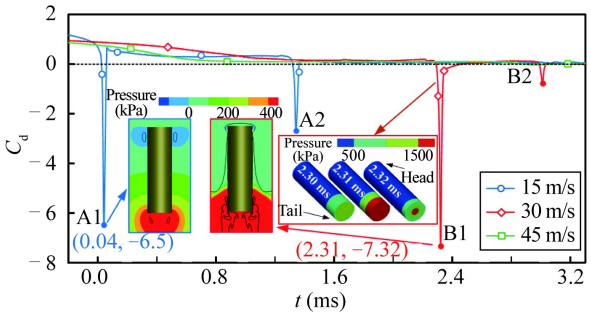
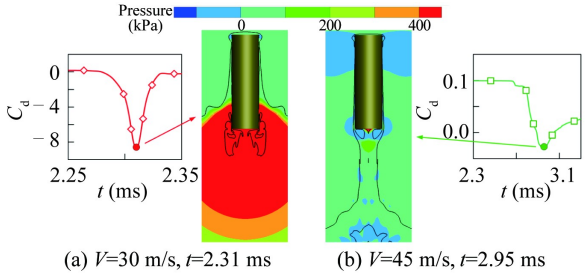


Fig. 13. Variation curve of the drag coefficient of cylinder at different initial velocities.

i.e. V=15 m/s and 30 m/s. This is because the partial cavity attached on the cylinder is unstable and vulnerable to collapse during water exit. In the case of collapse, transient high pressure is formed in the tail region of the cylinder (Point B1 at t=2.31 ms), leading to the negative instantaneous drag of the cylinder, that is an instantaneous thrust acting on the cylinder. There is no doubt that such instantaneous change of drag will cause non-negligible interference to the water exit stability of the cylinder.

It should be noted that the peak pressure occurs due to the collapse of shoulder cavity and acts on the side wall of the cylinder, as shown in Fig. 10, meaning that such peak pressure will only affect the lateral load and may cause damages on cylindrical surface. However, the drag coefficient peaks are dominated by the collapse of tail cavity behind the cylinder and the pressure is acting at the tail end of the cylinder, as shown in Fig. 13. Since the cavity around the cylinder at V=45 m/s just covers the whole cylinder and becomes a supercavity, the collapse of shoulder cavity occurs at the side end of the cylinder, while the relative stable tail cavity collapses quite weak and causes little influence on the cylinder. Although the cavity around the cylinder also shrinks and collapses to some degree, the pressure distribution around the cylinder is relatively uniform without significant pressure fluctuation (Fig. 9b3), therefore the change of drag coefficient during water exit is very gentle. As detailed in Fig. 14, there is also a negative peak of coefficient at V=45 m/s, with a much lower peak compared with that at V=30 m/s.



(a) V=30 m/s, t=2.31 ms (b) V=45 m/s, t=2.95 ms

Fig. 14. Drag peak of the cylinder and corresponding pressure distribution behind the cylinder.

4 Conclusions

In this paper, the VOF and overset grid technology are used to numerically simulate the cylinder's water exit process, and the characteristics of cavity evolution and forces acting on the cylinder and the surrounding flow field during the water exit process are detailed and discussed. The CFD framework is verified by mesh convergence analysis and a water-entry experiment of the same cylinder is conducted to validate the reliability of the present numerical model.

It is found that the wake region continues to expand as the cylinder moves upwards during water exit. The velocity distribution gradient tends to flatten gradually or even disappear as the cylinder moves away from the free surface, with the main influenced region being the cavity shedding region directly behind the cylinder. When the cylinder head passes through the free surface, there is an obvious pressure peak at the tail end of the cavity around the cylinder, and the higher the velocity, the greater the pressure. The tail pressure is affected by cavity collapse and the more severe the cavity collapse, the greater the tail impact pressure. These instantaneous pressure fluctuations may lead to the structural damage of the cylinder.

Considering the instability caused by collapse and shedding of tail cavity, when the cylinder exits the water with a partial cavity, there is an obvious instantaneous drag peak in negative direction, which is vulnerable to cause a sudden change of the cylinder's state and even motion instability. However, when the cylinder exits the water enveloped by a supercavity, the surrounding water region is relatively uniform and the drag changes gently, and thus the cylinder has good motion stability. It therefore can be concluded that the collapse of shoulder cavity mainly influences the load on the cylindrical surface while the tail cavity usually dominates the effects on the drag and motion stability during water exit.

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